

From Chatbots to **Agentic Workflows**

Responsible LLM Implementation in Academic Biostatistics

FRAME → SPECIFY → EXECUTE → VERIFY → INTERPRET → APPROVE

Steven C. Grambow, PhD

Associate Professor and Vice Chair of Education
Department of Biostatistics and Bioinformatics
Duke University School of Medicine

Disclosure, Transparency, and AI Use

Transparency is part of responsible AI practice.

Professional Disclosures

I serve on Data Monitoring Committees through WCG Consulting and for Gilead Sciences, Inc.

AI Use

Generative AI tools were used extensively during the development of this presentation.

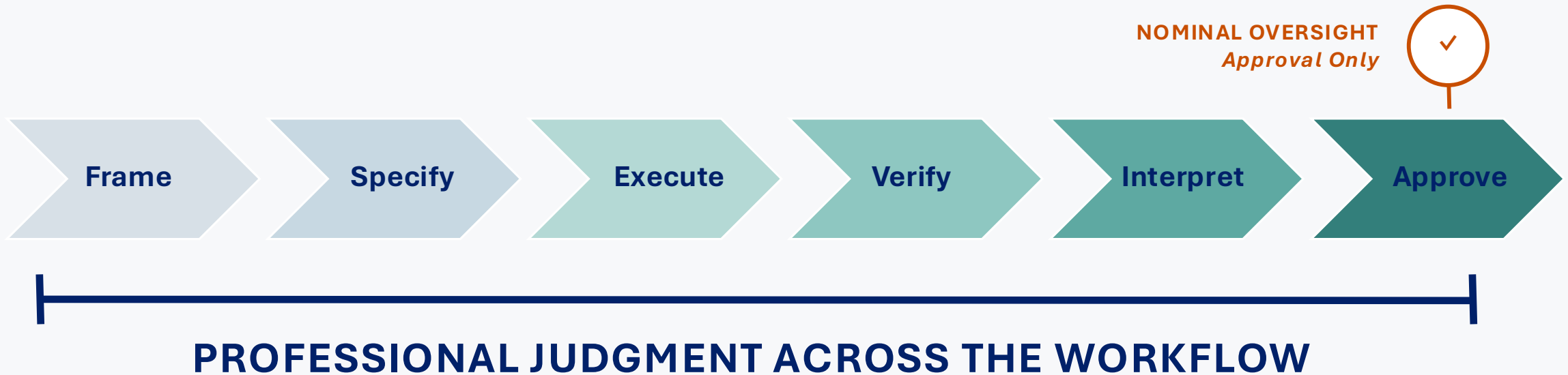
Responsibility

All content, interpretations, conclusions, examples, and recommendations were reviewed, revised, and approved by me.

I remain responsible for the final content and its interpretation.

AI contributed to the process. Professional responsibility remains human.

Human oversight is not the same as human judgment.



A person can approve the final output without having framed, challenged, or interpreted the work.

Integrating large language models in biostatistical workflows for clinical and translational research

Steven C. Grambow¹ , Manisha Desai² , Kevin P. Weinfurt³ , Christopher
J. Lindsell¹ , Michael J. Pencina¹, Lacey Rende¹  and Gina-Maria Pomann¹ 

Survey: September/October 2024

Adoption outpaced organizational readiness.

64%
(44/69)

Reported use

71%
(29/41)

Potentially consequential
errors

19% EACH
(13/69)

Formal Guidelines
Formal Training

ONGOING FOLLOW-UP

Survey currently underway

Approximately 10 academic medical centers

AI Adoption • Workflow Integration • Verification Practices • Organizational Readiness

Eight principles for responsible LLM use in biostatistical workflows.

1. Critical Evaluation

Assess outputs using domain expertise

2. Collaborative Verification

Engage colleagues to cross-check outputs

3. Multiple Tool Integration

Combine approaches for comprehensive checking

4. Continuous Learning

Augment knowledge; preserve critical expertise

5. Ethical Considerations

Align with guidelines, privacy, and bias standards

6. Transparency

Document LLM use and verification processes

7. Experimentation

Iterate on prompting to improve output quality

8. Stay Informed

Monitor model developments and best practices

Grambow et al., Journal of Clinical and Translational Science, 2025 May; 9(1):e131 · doi: 10.1017/cts.2025.10064

Judgment by Design

Four conditions make responsible judgment possible in practice.



Embedding responsible practice in the working repository

A teaching field test embedded the four design conditions in the files and procedures used throughout the project.

github/workflows	Add references appendix, switch to docs/ for GitHub Page...	2
R	Add simulation fidelity assessment and Critical Appraisal a...	2
docs	ci: re-render book [skip ci]	27
presentation-slides	docs: pre-conference repository updates, licensing, and s...	29
renv	Add renv scaffold: R 4.6.1, renv 1.2.3	2
source	Add IVAM-ED external chat transcript	
standards	Add Quarto book project: index.qmd and book structure	2
.Rprofile		2
.gitattributes		2
.gitignore		2
AGENTS.md		9
BOOTSTRAP.md		2
LICENSE.md		9
README.md		9
SECURITY.md		9
SESSION_LOG.md		9

Rendered Teaching Resource

IVAM-ED: A CRP 241 Teaching Case
Case Selection, Source Materials, Repository Design, and Learning Objectives

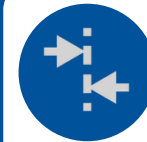
AUTHOR
Steve Grambow, PhD — Duke University

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About This Book

This book is a complete, reproducible tutorial in introductory clinical research methods built around a single published randomized trial. Every chapter uses the same dataset and the same study, so statistical concepts accumulate rather than restart. By the end you will have followed a real clinical investigation from raw data structure to published conclusion — examining the design, the analysis, the missing data, and the gap between what was planned and what was reported.

The primary language is R. All analysis is performed on a **synthetic (simulated) dataset** that approximates the published trial's sample characteristics. No real participant data are used or stored anywhere in this repository.



BOUNDARIES

AGENTS.md + standards/data-handling.md

Defines the data tier, permitted AI uses, approval gates, and prohibited actions.



LAYERED VERIFICATION

Executable checks + rendered outputs + subsequent review

Runs code, checks rendering, and compares simulated results with published targets.



PROCESS PROVENANCE

SESSION_LOG.md + simulation-refinement record

Records assumptions, identified problems, corrections, decisions, and unresolved limitations.



PROFESSIONAL CAPACITY

Instructor review + release authority

Sets educational goals, judges statistical fidelity, and retains responsibility for interpretation and publication.

Responsible implementation requires deliberate allocation and continued learning.

Allocate human involvement at the task level

HUMAN OUT

Bounded operations

Routine, constrained, reversible work with defined outputs and validation

HUMAN ON

Monitored execution

Work that can proceed under observation with clear intervention triggers.

HUMAN IN

Scientific Judgment

Framing, interpretation, consequential decisions, and final responsibility.

Automated checks can expand verification coverage. Professional judgment determines whether verification is sufficiently independent and proportionate to the consequence.

Challenge assumptions · Test code and outputs · Compare specifications with implementation · Identify unsupported claims

ORGANIZATIONAL LEARNING CYCLE

IMPLEMENT



OBSERVE



EVALUATE



ADAPT

Evidence from practice should inform revisions to boundaries, verification, process records, and professional development.